BRISBANE AIRPORT CORPORATION

MINIMUM DEVELOPMENT LEVEL CERTIFICATE

Area 51 Skygate

SECTION A – PROPERTY INFORMATION

DEVELOPMENT SITE NAME: Area 51

DEVELOPMENT SITE ADDRESS: SKG26, Skygate, The Blvd, Brisbane Airport QLD 4008

PRECINCT: Skygate

BUILDING USE: Amenities

DATE CERTIFICATE GENERATED: 2/01/2025

SECTION B - MINIMUM DEVELOPMENT LEVEL

MINIMUM DEVELOPMENT LEVEL is: 4.7 m Aerodrome Datum (AD)

Model Platform used for report: TUFLOW (Refer to **Figure 3** for model results extract)

SECTION C – DATA SUMMARY

MINIMUM DEVELOPMENT LEVEL Reference: Brisbane Airport Master Drainage Study (MDS) 2022

CATCHMENT: Battery Drain

DESIGNATED FLOOD EVENT FOR SITE is: 1% Annual Exceedance Probability (AEP) localised rainfall coinciding with the non-critical duration 1% AEP tailwater levels from Kedron Brook floodway via Battery Drain

SECTION D - NOTES

The additional information as described below is provided to qualify the Minimum Development Level (MDL) for the site. Please ensure that this advice remains complete when on forwarding.

The information supplied in this document represents the most current flooding information held by the Brisbane Airport Corporation Pty Ltd (BAC) at the time the certificate was generated.

The following should be noted when reading this Certificate:

- The above MDL is to be read in conjunction with the Flood Immunity Design Guidelines as shown in **Attachment 1**;
- The flood model extent and depths shown in **Attachment 2** should be used as a guide to appreciate the overall flood risk across and outside the site boundary including access roads.
- For large lots, where there is over 500mm difference in 1% AEP flood levels within the site, the designer may consider different development levels by meeting the criteria specified in **Section E**.
- Brisbane Airport Master Drainage Study 2022 must be referenced for detailed information about the design flood events
- Levels of Feb 2022 historic event are based on the TUFLOW model run of the actual flood event informed by data gathered/surveyed during 2022 February flooding at the Brisbane Airport.
- BAC provides the AD to AHD conversion for information purposes. Please note that all documents submitted to BAC shall be referenced to AD on airport (refer to **Attachment 3**);
- The developer shall be responsible for determining the Finished Surface Level of the development;

- The Finished Floor Level (FFL) shall incorporate an appropriate settlement rate for the site;
- All water treatment processes shall be carried out on the site before discharge off-site; and
- On-site water treatment processes shall be carried out in accordance with BAC's Stormwater Quality Management Guidelines.

Table 1 below provides the design criteria for the different applications located on airport:

Table 1: BAC Flood Immunity Design Criteria

Development	Land use	Criterion	Flood immunity (% AEP)
	Primary Criteria		
Pavement	Runways	No ponding (including shoulders)	0.2%
Pavement	Taxiways	No ponding (including shoulders)	0.2%
Pavement	Aprons	No ponding (including shoulders)	1%
Buildings and Development Areas	Terminal and Operational	No ponding within 0.3m of floor level	0.2%
Buildings and Development Areas	Public Transport	No Ponding	1%
Buildings and Development Areas	Airport Access Road	No ponding	2%
Buildings and Development Areas	Active Transport	No ponding	2%
Buildings and Development Areas	Other Roads and Car Parks	No ponding	2%
Buildings and Development Areas	Undercroft carparks		1%
Buildings and Development Areas	Commercial/Industrial	No ponding within 0.3m of floor level	1%
Buildings and Development Areas	Other	No ponding within 0.3m of floor level	2%
	Secondary Criteria		
Grassed Areas	Greater Runway Strip	Ponding within 75m of Runway centreline not to exceed 12hrs duration	18%
Grassed Areas	Taxiway and Apron Flanks	Ponding within 15m of pavement edge not to exceed 12hrs duration	18%
Glide Path Localiser	Primary Portions	No ponding	2%
Glide Path Localiser	Remainder	Ponding not to exceed 12 hrs duration	18%

Table 2: Comparing AEP and ARI

Annual Exceedance Probability (AEP)	Average Recurrence Interval (ARI) Years
63%	1
39%	2
18%	5
10%	10
5%	20
2%	50
1%	100
0.5%	200
0.2%	500

SECTION E – DESIGN RESPONSIBILITIES

The designer will be required to submit to BAC an analysis demonstrating no worsening of risk to adjoining sites from the development by detailing as a minimum:

- Major event flood levels ie 1% AEP storm tide or 1% AEP Brisbane River/Kedron Brook as applicable (whichever is highest) combined with 10% AEP local rainfall
- 2% AEP local event runoff with HAT as the tailwater
- 10% AEP local event runoff with HAT as the tailwater
- Water quality measures adopted to meet the requirements of *the Airports (Environment Protection) Regulations 1997.*

It is the responsibility of the designer to determine the level of risk chosen which governs the building pad level or levels of any other critical infrastructure placed on the site. For example, the designer may choose to include settlement on top of the immunity flood level; however, the settlement incurred during ground improvement works must not be lower in its final form than the provided immunity level for the chosen Finished Floor Level or infrastructure level.

Direct flood model results of all simulated design flood events ie 10% AEP, 2% AEP and 1% AEP will require appropriate freeboard allowances and climate change considerations commensurate to the use of the development/infrastructure, design life and planning horizon. Flood level information shown in **Attachment 2** and climate change scenarios available in Brisbane Airport Master Drainage Study 2022 (updated 2024 as part of Terminal 3 Precinct Planning) may be used as a guide for this purpose.

Important things you should know about this MDL certificate including the Figures.

Third Parties

- It is not possible to make a proper assessment of the drawings/figures and associated concepts without an investigation and clear understanding of the terms of engagement under which the MDL has been prepared, including the scope of the instructions and directions given to and the assumptions made in preparing this MDL certificate.

- The Figures and associated concepts may not address issues which would need to be addressed with a third party if that party's particular circumstances, requirements and experience with such assessments were known and may make assumptions about matters of which a third party is not aware.

- BAC therefore does not assume responsibility for the use of, or reliance on the drawing and associated concepts by any third party and the use of, or reliance on, the MDL certificate by any third party is at the risk of that party.

Existing Property: Inherent Risk

- The owner or prospective purchaser of an existing property or asset necessarily assumes the risk of there being defects inherent in the asset. An engineer's report can assist an owner, lessee or prospective purchaser in making an assessment of that risk but does not eliminate that risk.

- A drawing and associated concept of this nature is not a certification, warranty or guarantee.

Further enquiries regarding this document should be directed to:

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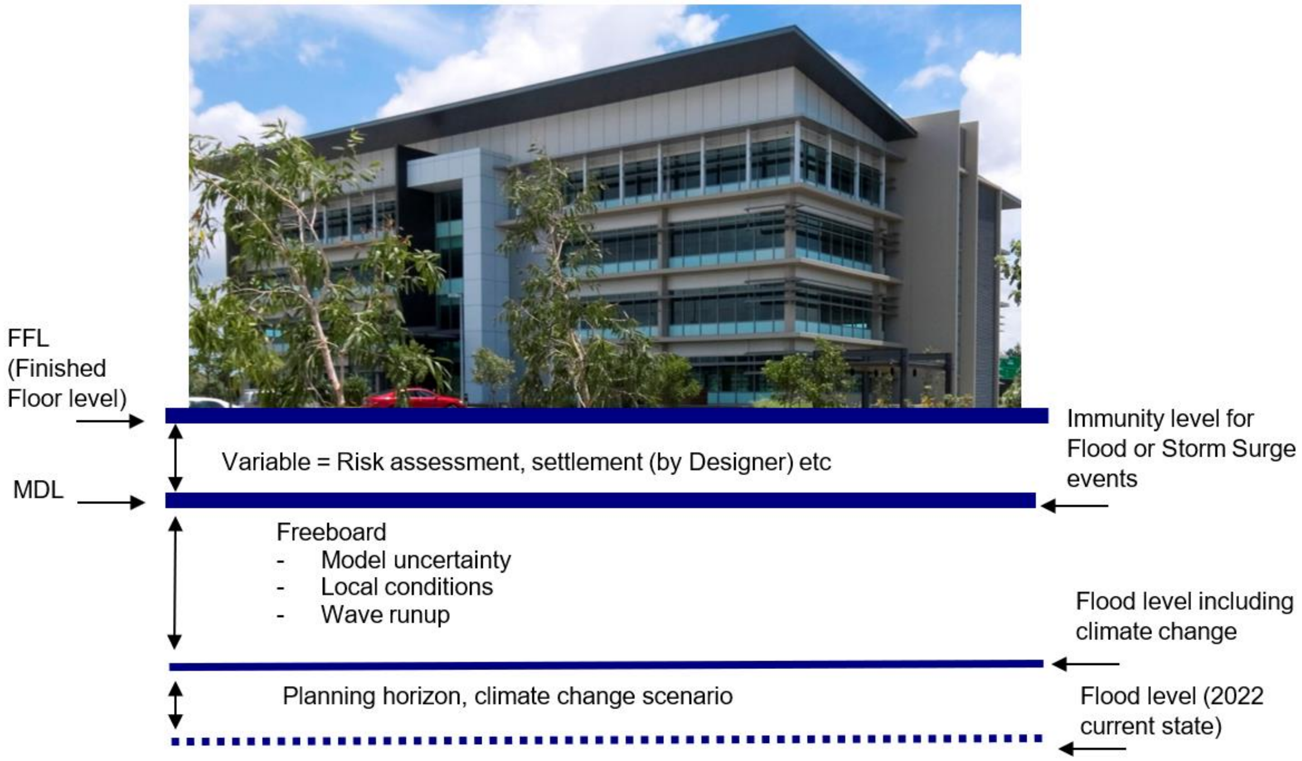
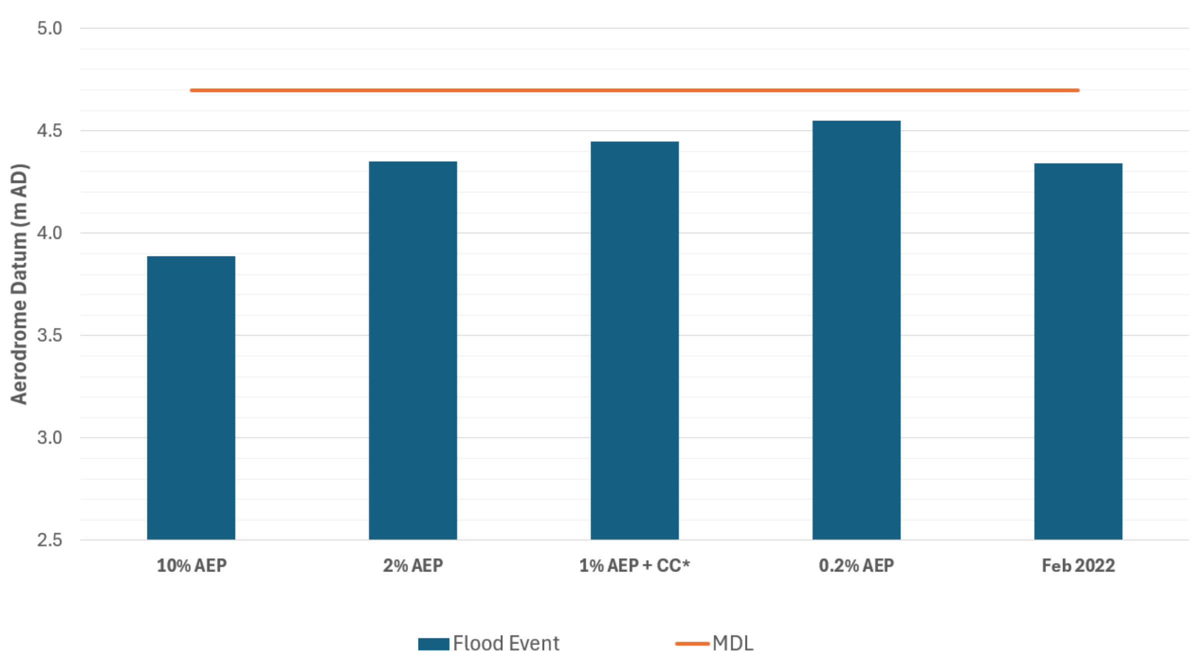
ATTACHMENT 1

Figure 1 - Earthworks and Drainage Design Guidelines – Minimum Development Level and Finished Floor Level



ATTACHMENT 2

Figure 2 – Flood modelling information



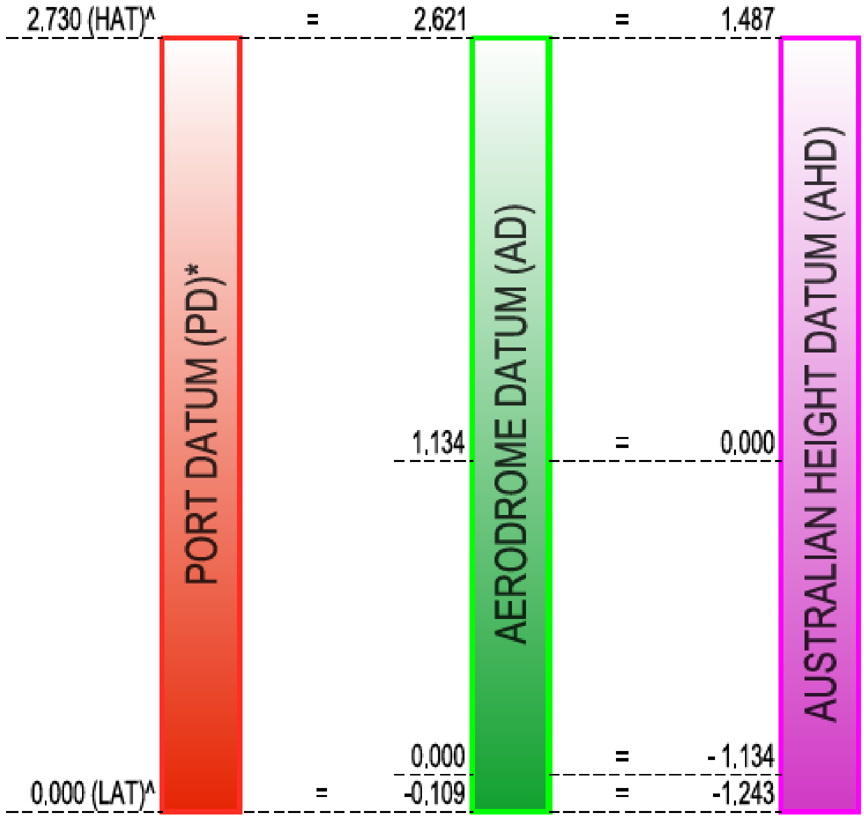
CC* *- MDS simulated climate change scenario at 2100 Planning Horizon (SSP5-8.5) in line with IPCC Report 6 Shared Socioeconomic Pathways*

Figure 3 - Flood extent of 1% AEP event with Climate Change (2022 MDS)

ATTACHMENT 3

AD TO AHD CONVERSION

AERODROME DATUM RELATIVE TO AUSTRALIAN HEIGHT DATUM & PORT DATUM



**NOTE: THE TIDAL DATUM VARIES FROM PLACE TO PLACE AND THE ABOVE ILLUSTRATION IS CORRECT FOR THE BRISBANE BAR. HEIGHTS ARE RELATIVE TO MEAN SEA LEVEL FOR THE LOCATION AND NOT TO AHD.*
^HAT: HIGHEST ASTRONOMICAL TIDE; LAT: LOWEST ASTRONOMICAL TIDE

CONVERSIONS:

TO CONVERT LEVELS ON PD TO AD, SUBTRACT 0.109

TO CONVERT LEVELS ON PD TO AHD, SUBTRACT 1.243

TO CONVERT LEVELS ON AD TO AHD, SUBTRACT 1.134

TO CONVERT LEVELS ON AD TO PD, ADD 0.109

TO CONVERT LEVELS ON AHD TO AD, ADD 1.134

TO CONVERT LEVELS ON AHD TO PD, ADD 1.243